CSE 2215: Data Structures and Algorithms - I , Summer 2023

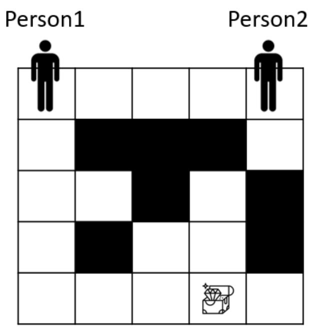
Class Test 5

Set A

Marks : 20 **Time: 20 min**

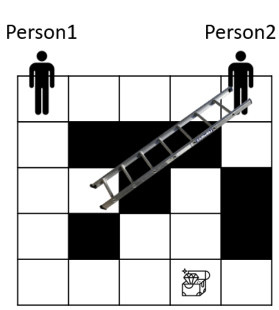
1 Consider the given **maze** with several square shaped tiles. Draw a graph from the maze based on the following hints. **[5]**

I. You **can move** from one maze to another either **vertically or horizontally**.
II. You **cannot** move to a maze which has a **wall** inside it (marked with **black** color).
III. While drawing a graph, form a vertex **for each maze without a wall.**
IV. If a person can move from one maze to another, draw an edge connecting its corresponding vertices.



2. Using the graph from question 1, calculate the distance between the **Person1** and the treasure and after that calculate the distance between the **Person2** and the treasure. [Hint : Use BFS/DFS. Once for Person1 and once for Person 2] **[5]**

3. Consider that there is a bridge connecting the position of Person2 with the position shown in the diagram. Redraw the graph again. Can Person2 reach the treasure before Person1 this time? **[5+5]**



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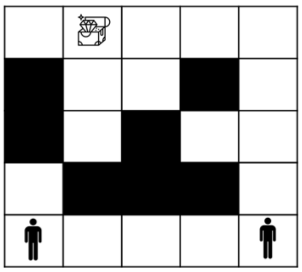
Class Test 5

Set B

Marks : 20 **Time: 20 min**

1 Consider the given **maze** with several square shaped tiles. Draw a graph from the maze based on the following hints. **5**

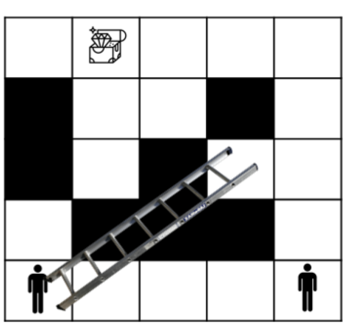
V. You **can move** from one maze to another either **vertically or horizontally**.
VI. You **cannot** move to a maze which has a **wall** inside it (marked with **black** color).
VII. While drawing a graph, form a vertex **for each maze without a wall.**
VIII. If a person can move from one maze to another, draw an edge connecting its corresponding vertices.



Person2 Person1

2. Using the graph from question 1, calculate the distance between the **Person1** and the treasure and after that calculate the distance between the **Person2** and the treasure. [Hint : Use BFS/DFS. Once for Person1 and once for Person 2] **5**

3. Consider that there is a bridge connecting the position of Person2 with the position shown in the diagram. Redraw the graph again. Can Person2 reach the treasure before Person1 this time? **5+5**



Person2 Person1